METRIC

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SECTION 533
22 May 1978

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MILITARY STANDARD

INTERFACE STANDARD FOR SHIPBOARD SYSTEMS

SECTION 533

POTABLE WATER SERVICE (*METRIC*)

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DOD-STD-1399(NAVY)
SECTION 533
22 May 1978

DEPARTMENT OF THE NAVY

WASHINGTON, DC 20362

Interface Standard for Shipboard Systems,
Potable Water Service

DOD-STD-1399(NAVY)
SECTION 533

1. This Military Standard is approved for use by all interested Commands of the Department of the Navy in the technical development plans, design, and procurement specifications for new ship acquisitions, ship modernizations or conversions, and equipment for installation therein and into active fleet ships where applicable, and is available for use by all Departments and Agencies of the Department of Defense.

2. Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Ship Engineering Center, SEC 6124, Department of the Navy, Washington, DC 20362, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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FOREWORD

Purpose. This section defines the standard interface requirements for, and the constraints on, the design of shipboard equipment which will utilize the shipboard potable water service.

Nature of the interface. Navy ships have a requirement for a shipboard potable water service system supplying a variety of uses and at differing pressures and temperatures. This section covers the shipboard potable water service system and is concerned with the delivery of an adequate supply of potable water of satisfactory quality to various users and to other independent systems and subsystems.

Structure. The technical content first delineates the characteristics of the shipboard potable water service system in terms of water quality, pressure and temperature. The constraints on equipment design and installation which are necessary to achieve shipboard compatibility with these characteristics are then established.

Numerical quantities. Numerical quantities stated in this section are expressed in metric (SI) units and may be followed by inch-pound units in parentheses. The SI equivalents of the inch-pound units are approximated to a practical number of significant figures. Where parenthetical values are given in inch-pound units, the inch-pound units are to be regarded as the current specified magnitude.

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1. GENERAL, SCOPE, INTERFACE, AND APPLICABILITY

1.1 General. The policies and procedures established by DOD-STD-1399 are mandatory. This section and the basic standard are to be viewed as an integral single document.

1.2 Scope. This section establishes interface requirements for shipboard equipment utilizing potable water services to ensure compatibility between such equipment and the potable water service system.

1.3 Interface. The basic characteristic and constraint categories concerned with this interface are shown symbolically on figure 1 (see section 3, "Definitions" of DOD-STD-1399):

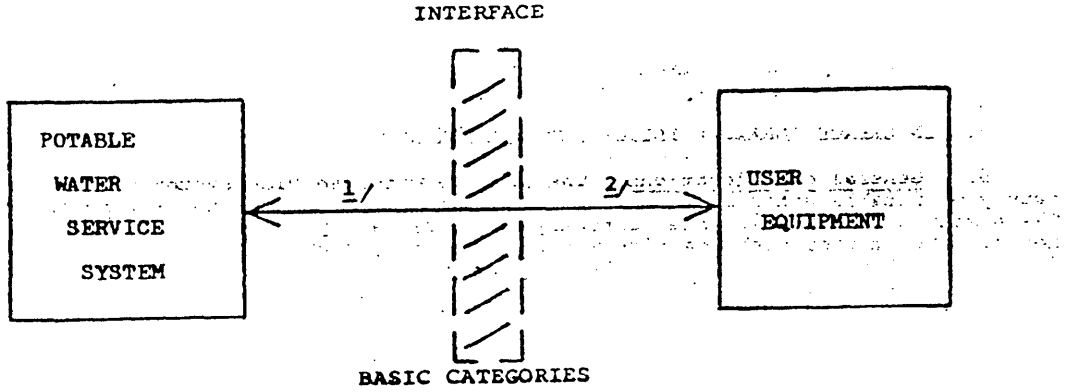


FIGURE 1. Interface.

The particular interface characteristics and constraints pertinent to this section are described in 5.2 and 5.3.

1.4 Applicability. This section is directly concerned with the design and installation considerations at the interface between the potable water system and its users which affect the overall compatibility at the interface. For criteria concerning related fresh water interface areas, see other pertinent sections of DOD-STD-1399 in the 530 series.

2. REFERENCED DOCUMENTS

2.1 Issues of documents. The following document of the issue in effect on the date of invitation for bids or request for proposal forms a part of this standard to the extent specified herein.

PUBLICATION

MILITARY

BUREAU OF MEDICINE AND SURGERY (BUMED)
NAVMED P-5010 - Manual of Naval Preventive Medicine.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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3. DEFINITIONS

3.1 Potable water. Potable water is water whose bacteriological quality and physical and chemical characteristics make it suitable for human consumption.

3.2 Potable water service system. A potable water service system is a shipboard system, including the necessary tank filling, storage, transfer, and service arrangements for the distribution of potable water throughout the ship.

3.3 User equipment. A user equipment is any facility or apparatus which utilizes or consumes water from the potable water service system.

4. REQUIREMENTS

4.1 Requirements. The specific interface requirements and constraints established herein are mandatory and shall be adhered to by SYSCOMs, Project managers, contractors, and all others engaged in any aspect of shipboard potable water service system design to which these requirements and constraints apply, including equipment design, production, and installation (see section 4, "Requirements" of DOD-STD-1399).

5. INTERFACE CHARACTERISTICS AND CONSTRAINTS

5.1 General considerations. The potable water service system provides for a continuous supply of potable water from the storage tanks to the various users throughout the ship. The system provides cold water directly to users, and provides water to heaters for hot water needs. The storage tanks receive water from shore, from another ship or from the ship's distilling plants. Except on nuclear submarines, the water will be disinfected by bromine disinfection systems. Figure 2 shows the basic elements of a typical potable water service system.

5.2 Interface characteristics. The interface characteristics of the ship's potable water service system are described in 5.2.1 through 5.2.4.

5.2.1 Water quality standards. When water enters the potable water service system, it is assumed to be of a satisfactory quality with respect to bacteriological content and physical and chemical characteristics. Water quality standards and the criteria for their application are given in NAVMED P-5010. Detailed water quality standards are provided in the Appendix for information.

5.2.2 Pressure. The water pressure supplied will be within the range of 35 kilopascals (kPa) (5 pounds per square inch (lb/in^2)) to 700 kPa (100 lb/in^2).

5.2.3 Temperature. The temperature of the water supplied will be as follows:

(a) Cold potable water - 2°C to 32°C (35°F to 90°F).
(b) Hot potable water - 52°C to 57°C (125°F to 135°F).

5.2.4 Bromine chlorine residual. A minimum measurable free bromine or chlorine residual of 0.2 milligrams per liter (mg/L) will be maintained in the potable water distribution system at all times, except on nuclear submarines (see 20.5 of the Appendix).

5.3 Interface constraints. The interface characteristics of the ship's potable water system impose certain constraints on the design and installation of equipment utilizing or consuming potable water. These constraints are specified in 5.3.1 through 5.3.3.

5.3.1 Compatibility. The design and installation of equipment utilizing potable water service shall be compatibile with the interface characteristics given in 5.2.

5.3.2 Integrity of the potable water system at a delivery point. When hot or cold potable water is delivered to user equipment, the potable water system shall be protected from contamination.

5.3.2.1 Air gaps. Where potable water is delivered to other systems, tanks, or facilities that could contain water of inferior quality, an air gap of at least two supply pipe diameters shall be provided between the supply and receiving connections. A funnel shall be provided in the air gap receiving connection wherever necessary. Location of faucets above the rims of lavatories, sinks, and other open receiving vessels constitutes an adequate

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air gap. Some examples of facilities which require an air gap are: galley and pantry sinks, ice machines, dishwashers and other scullery equipment, vegetable peelers, steam kettles, laundry machines, and sterilizers.

5.3.2.2 Hose connections. Where potable water is delivered to a user equipment by means of a hose connection, the connection shall be fitted with a valve of the stop-check type and a vacuum breaker back flow preventer.

5.3.3 External filling and delivery connections. The connections for receiving potable water from external sources, or delivering water to another ship shall be:

(a) Surface ships: 63 millimeters (mm) (2-1/2 inch) hose valve connection (except for ships with a rated pump capacity of less than 50 gallons per minute, where the hose valve connection shall be 38 mm (1-1/2 inch) located on the weather deck.
(b) Submarines: 38 mm (1-1/2 inch) hose valve connection located within the pressure hull.

Such hose valve connections shall be painted dark blue, provided with a cap and chain, and plainly marked "POTABLE WATER ONLY".

6. DEVIATIONS

6.1 Conditions. In achieving the purpose of this section, it is recognized that there must be some flexibility of application. During the early design stage of shipboard equipment utilizing potable water service, it may become apparent that significant advantages in the overall design/operation of such equipment can be achieved by deviating from the standard characteristics specified herein. In such instance, the provisions of section 6, "Deviations" of DOD-STD-1399 shall be complied with.

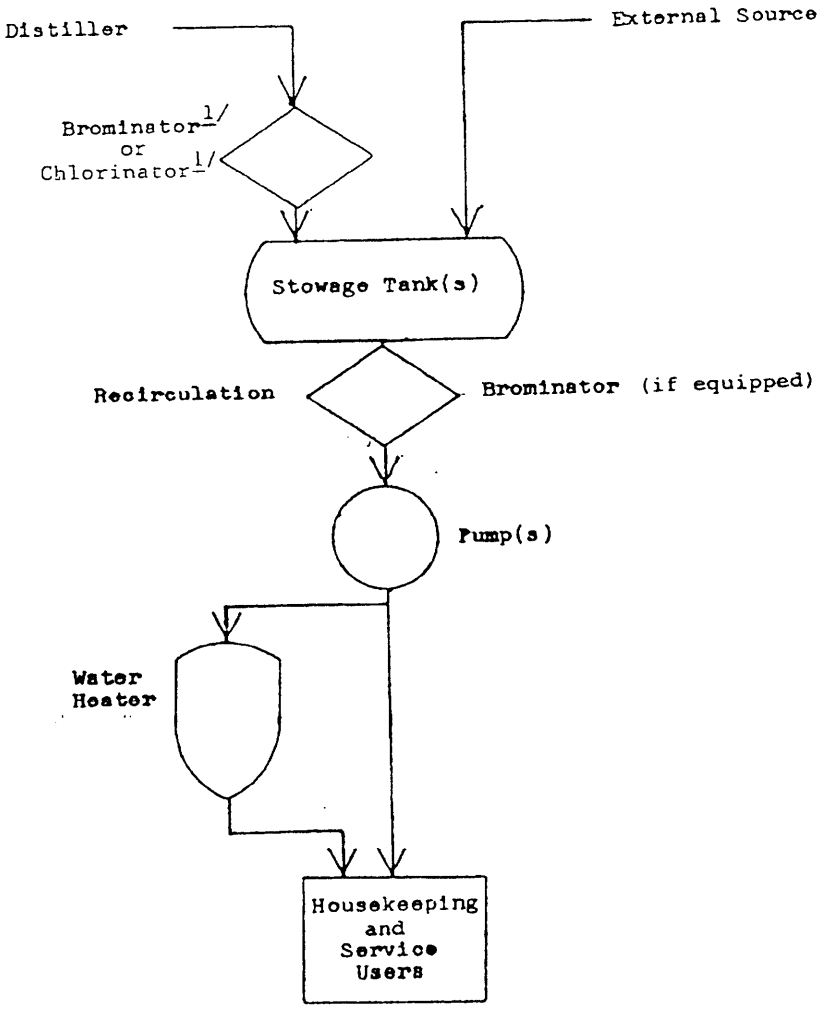
6.1.1 Deviation procedure. Requests for deviations shall be submitted to the Naval Sea Systems Command with copies to:

(a) Program/Project manager.
(b) NAVSEC 6154.

Review activity:
YD
User activities:
EC, AS

Preparing activity:
Navy - SH
(Project 1990-0022)

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1/ Not used on nuclear submarines

FIGURE 2. Basic elements, typical potable water service system.

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APPENDIX

WATER QUALITY STANDARDS

10. APPLICATION

10.1 The material in this appendix is provided for general information and guidance to assist the designer in complying with the requirements of this section.

20. WATER QUALITY STANDARDS

20.1 General. Potable water entering the service system is assumed to be of satisfactory quality as prescribed in NAVMED P-5010. The prescribed standards are summarized in 20.2 through 20.5.

20.2 Bacteriological quality. Determination of the bacteriological quality of potable water is concerned with the presence of the indicator coliform group. If water sampled in accordance with the prescribed procedures shows the presence of these organisms above the level prescribed in NAVMED P-5010, it is cause for the rejection of the water.

20.3 Physical characteristics. While color and threshold odor are characteristics of aesthetic importance, turbidity is one of significant health impact due to its potential interference with proper disinfection and microbiological analysis. The limits prescribed are:

(a) Turbidity:
  (1) 1 unit.
  (2) Up to 5 units if it can be demonstrated that the turbidity does not interfere with disinfection or interfere with microbiological determinations.
(b) Color - 15 units.
(c) Threshold odor number - 3.

20.4 Chemical characteristics. Chemical characteristics of potable water are determined by the level mg/L of various chemicals in solution. Table I notes maximum contaminant levels (MCLs) of chemicals specified in current Environmental Protection Agency National Primary Drinking Water Regulations. MCLs for additional inorganic and organic compounds will be added when toxicological data becomes available.

TABLE I. Chemical characteristics.

Chemical	Maximum contaminant level (mg/L)
Arsenic	1.05
Barium	1
Cadmium	0.01
Chromium	.05
Lead	.05
Mercury	.002
Nitrate (as nitrogen)	10
Selenium	0.01
Silver	.05
Endoin	.0002
Lindane	.004
Methoxycholr	.1
Toxaphene	.005
2,4 D	.1
2,4,5-TP Silvex	.01

20.4.1 Fluorides. The fluoride level is not critical in shipboard potable water (see NAVMED P-5010).

20.5 Bromine/chlorine residual. A minimum measurable free bromine or chlorine residual shall be maintained in the shipboard potable water distribution system at all times except on nuclear submarines. Brominators are being installed on Navy surface ships, but older ships may still rely on chlorination, in which case they will maintain a minimum measurable

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3.1 Compartment ambient [illegible] ambient conditions are

chlorine residual. Since potable water from distillers, and at times water from external sources, will not contain the prescribed residual, it will often be necessary to treat the water to bring it into conformity with this requirement. The residual should not go below 0.2 mg/L for either bromine or chlorine.

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